



YASSINE EL HOUM

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● ABOUT ME

Researcher - Control systems

Social media : [LinkedIn profile](#) , [GitHub profile](#)

● WORK EXPERIENCE

RESEARCH CONTRACT – NATIONAL CENTER FOR SCIENTIFIC AND TECHNICAL RESEARCH IN RABAT (CNRST) –
01/09/2017 – 01/06/2021 – RABAT , MOROCCO

Project: Design and implementation of an experimental platform for the study, construction, and optimization of solar car prototypes

Tasks:

1. Build a sun simulator capable of simulating natural sunlight conditions.
2. Construct a robust performance tracking system based on NI DAQ 6321 that tracks, monitors, and analyzes the performance of the solar vehicle.

Objectives:

1. Evaluation of the electrical power generated by the panels.
2. Assessment of efficiency.
3. Verification of the condition of solar panels and detection of defective PV panels

Accomplishments:

1. Participation and securing a podium position in the Moroccan Solar Race Challenge 2019, an international event organized by the Research Institute on Solar Energy and New Energies (IRESEN) in Morocco
2. Invention patent, URL Link « [Technical inspection station for electric and solar vehicles](#) »

Technical Environment : Matlab Simulink, NI LabVIEW/NI DAQ 6321

Technical skills acquired:

Hardware targets used:

- Texas Instruments C2000 Family: F28x F2812, and C2000 Delfino F28335, F28379D
- National instruments data acquisition DAQ PCIe 6321 (x series) & BNC2110
- dSPACE 1104 Control Card
- OPAL RT Real time target machine

Software and Programming:

- Matlab/Simulink, NI LabVIEW, ControlDesk, PSIM.
- C, C++, Python

Estimation/Filtering:

- Estimation of the position and velocity of a BLDC motor based on sliding mode observer.

Control /Identification:

- Robust control of three phase machines, particularly BLDC motors used in solar vehicles
- Optimization of photovoltaic energy using MPPT control techniques

Embedded Electronics:

- Development of a dashboard for real time data analysis in a solar vehicle
- Communication bus SCI, SPI, I2C, CAN, MAVLink

RESEARCHER - ELECTRICAL ENGINEERING AND ADVANCED CONTROL – EMI – 01/05/2023 – RABAT, MOROCCO

Research activities

1. Improvement of trajectory tracking for an aerial vehicle in a disturbed environment.
2. Contribution to robust control laws based on advanced sliding mode theory.
3. Implementation of control laws in an embedded system for a quadrotor.
4. Comparative study with solutions proposed in the literature

Research activities (with the team)

1. Hardware implementation of robust control laws on the DSP F28335 board to regulate the speed of a BLDC (Outrunner) motor.
2. Extraction of optimal power from a photovoltaic installation using MPPT control techniques implemented on the DSP F2812 board.
3. Binary diagnosis of Hall effect sensors in BLDC motors

Supervision Activities:

1. Assistance of practical sessions for third year engineering students, specializing in electrical engineering
2. Supervision of mini projects for second year engineering students, specializing in electrical engineering.
3. Member of the organizing committee for scientific days at the EPC laboratory, held at the Mohammadia School of Engineering

Technical Environment : Matlab Simulink, dSPACE/ControlDesk, OPAL RT, TI C2000, PX4

Technical skills acquired:

Modeling:

- Systems dynamical modeling using Matlab/Simulink

Control /Identification:

- Optimal trajectory tracking via robust adaptive control based on sliding mode theory for aerial vehicles, especially quad-rotors
- Gray box system identification of an unmanned aerial vehicle, specifically a quad-copter

Estimation/Filtering:

- State estimation of aerial systems' position and orientation using Kalman filters

POSTDOCTORAL RESEARCHER – UNIVERSITY OF CATANIA – 23/09/2024 – Current – CATANIA, ITALY

Project: Implementation of a BCI Platform for Controlling Robotic Systems for Home Automation (PRIN PNRR 2022 – HOME4.0 Project).

Project Activities:

Brain–Computer Interface (BCI) System Development :

- Designed and implemented a real-time BCI pipeline for decoding EEG signals using *ROS-Neuro* and *BCI* hardware.
- Developed signal processing and feature extraction modules for motor imagery-based control.
- Integrated machine learning classifiers for translating brain signals into binary/intention commands.

Human–Robot Interaction for Home Automation:

- Implemented a closed-loop control interface enabling users to control assistive and service robots via BCI.
- implemented communication layers between the BCI and robotic systems based on *ROS*.
- Created autonomous behaviors and safety constraints for home-assistance scenarios.
- Built and tested simulation environments in Gazebo for robot navigation.
- Conducted tests scenarios for BCI-controlled movement and pick-and-place tasks in real environment (Alter-ego robot).

Technical Skills acquired:

- Brain–Computer Interface (BCI): EEG signal acquisition, preprocessing, feature extraction, classifier training.
- Robotics & Control: *ROS/ROS-Neuro*, Gazebo, MoveIt.
- Programming: Python, MATLAB/Simulink, C++, Bash scripting.
- Machine Learning: Classification (SVM, LDA), probabilistic models for decision-making under uncertainty.
- System Integration: Real-time communication (*ROS* topics/services), delay compensation, hardware–software synchronization.
- Software Tools and operating systems: Docker, Git, Ubuntu/Linux environment management.

Soft Skills

- Cross-disciplinary collaboration (Neuroscience, Robotics , Control).
- Project documentation and dissemination of research outcomes.
- Mentoring and collaboration with academic teams and students.

● EDUCATION AND TRAINING

01/06/2014 Settat, Morocco

BACHELOR OF SCIENCE AND TECHNOLOGY (LST) Faculty of Sciences and Techniques in Settat (FSTS)

Field of study Electronics and automation

01/06/2016 Settat, Morocco

MASTER OF SCIENCE AND TECHNOLOGY (MST) Faculty of Sciences and Techniques in Settat (FSTS)

Field of study Electronics and automation

Field of study Electronics and automation

LANGUAGE SKILLS

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
FRENCH	C2	C2	C1	C1	C1
ENGLISH	C1	C1	C1	C1	C1
ITALIAN	A2	B1	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Technical skills

modeling and simulation of dynamic systems | sensor fusion and state estimation | Git | Linux | Python | BCI
signal decoding and system integration | MATLAB/Simulink | ROS / ROS-Neuro | control theory (nonlinear, optimal,
and safety-constrained control) | Robotics and autonomous systems | embedded systems | ptimization and data-
driven control | Docker | PX4

PUBLICATIONS

2017 – CURRENT
SCIENTIFIC PUBLICATIONS

Google Scholar: [Publications](#)